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Statement of Purpose, UCSB CS Ph.D.

My research mainly focuses on developing more efficient and robust natural language processing (NLP) models. Even though current deep learning approaches show remarkable performance in NLP, there is still much room for improvement, especially in terms of efficiency and robustness. First, due to their prohibitive computational cost, enhancing the efficiency of NLP models is crucial to use them in practice. Second, because the distribution of training data and test data often mismatch in real NLP applications, effective training methods to generalize well to unseen data are required. Through my graduate study, I want to explore themes including (1) memory augmentation, (2) efficient computation, and (3) robust training.

Memory Augmentation: I am excited about incorporating memory to language models for efficiently increasing model capacity or injecting knowledge information into the model. As initial progress, I successfully augmented product key memory (PKM) into pretrained language models with initialization and residual memory, achieving BERT-large performance with a PKM-augmented BERT-base while much faster [P#10]. However, in this approach, memory slots are hard to control, and the memory-augmented model is not parameter-efficient. I want to design memory interpretable and compact like Entities as Experts (EaE) as future work.

Instead of using memory as network parameters, other works save key-value pairs as non-parametric memory externally and use it by the retrieval. For instance, the nearest neighbor language modeling constructs a datastore (memory) for improving accuracy and generalization without additional training. Also, Dense-Sparse Phrase Index (DenSPI) is a memory storing vectors corresponding to phrases in a corpus and used to answer a question by querying from the index. There are many other examples, including REALM, DPR, RAG, and BLINK.

I will further investigate to find an effective solution to exploit memory. I have a deep interest in examining which method among parametric or non-parametric ways is better to use memory. Generative entity retrieval (GENRE) recently proposed from FAIR is extremely interesting for me in that this method reformulates entity linking to a conditional generation problem and removes the necessity of storing a non-parametric memory (dense representations of a large number of entities). As shown in recent works, both sides have their own benefit, so combining the best of both worlds in a unified system could be an interesting direction.

Efficient Computation: I am interested in improving the accuracy-efficiency trade-off in the transformer inference and designing better model architectures. To this end, I recently have proposed Length-Adaptive Transformer [P#12], trained with random length drop and used with a multi-objective evolutionary search on length configurations for anytime prediction. Alternative adaptive dimensions in addition to adaptive sequence length could be layer depth with the early-exit (e.g., DeeBERT) and size of the model architecture (e.g., DynaBERT). I want to figure out why adaptive computations could reasonably perform well and the outcome when those methods are combined. Also, I will extend token-level drop to span-level or sentence-level for much efficient length drop and application to tasks that require longer context. Another direction of an extension would be using a separate model to decide which and how many tokens to drop example-wise, that trained with reinforcement learning.

Beyond classification tasks, efficient sequence generation is more challenging as the search space of this structured prediction problem is exponentially large. Previously, I tried efficient generation in ASR error correction, query auto-completion [P#8], and dialog state tracking

[P#9]. I want to study more about efficient decoding methods for text generation, such as non-autoregressive generation, latent variable modeling, and finding optimal granularity.

Robust Training: I am generally interested in advanced training methods, such as semi-supervised learning (SSL) and self-training, that somehow utilize unlabeled data to overcome a limited amount of labeled data to cover all possible data distribution. The current convention in NLP is transfer learning from a pretrained language model followed by finetuning. The optimal pretraining strategy is still not fully explored yet, and my industrial experience in large-scale pretraining will help to enhance it further. Our LaRva (Language Representation by Clova) team in NAVER built a pipeline to pretrain large-scale language models for Korean and Japanese. Also, I worked on end-to-end spoken language understanding for speech-based dialog service. Considering that collection of speech data is more expensive than text data, leveraging textual information by cross-modal pretraining or cross-modal knowledge distillation was extremely helpful [P#13, P#14].

Besides, I am currently delving into data augmentation methods in NLP that can be used in the finetuning stage to improve generalization. Contrary to other approaches that apply generic methods to continuous embedding vectors, I consider discrete characteristics of text data. One is virtual adversarial training with discrete input perturbation by word replacement [P#15], and the other is applying a mixup with a span of input tokens [P#16]. Those methods have shown to be superior to the continuous counterpart and beneficial to SSL and even supervised learning in NLP.

Moreover, I wish to investigate self-training methods. I want to check one simple idea shortly that might be already studied by others. Related to the memory augmentation part, retrieved information from non-parametric memory (e.g., a datastore in the nearest neighbor language modeling) can help to generate a better prediction. We can use this better prediction as supervision by enforcing the prediction solely based on model parameters without memory close to it in the form of self-distillation. With improved model parameters, we can dump memory again and do this procedure iteratively until the prediction accuracy saturates.

Conclusion: As illustrated above, advancing the efficiency and robustness of NLP models makes me very excited. Over three years of experience in the industry encourage me to decide which problems are worth tackling and focus on practically essential research questions. I always raise a question about standard practices but non-trivial choices to make current systems better. I am a self-motivated independent researcher and enjoy collaboration with others.

I am about to start a new journey of more in-depth research. I am undoubtedly certain that pursuing a Ph.D. in UCSB CS would allow me to grow as an outstanding researcher and contribute to the community. Above all, I hope to collaborate with amazing people in the UCSB NLP group, including *William Wang* and *Xifeng Yan*. I am fascinated by their research and believe that their research interests fit well with mine. I am looking forward to doing compelling research in the excellent research environment of UCSB. I also aspire to keep my academic career as a professor after graduation. Based on my mentoring experience with many interns, I find that I love teaching and advising as well as my own research.

♠ [P#X] corresponds to the X-th paper in the publication list of my CV.